Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 6

Biomedical Ethics

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Learning Objectives

1. Identify & explain the basic principles that govern Human Subjects research
2. Identify the types of information that must be provided to get approval from an Institutional Review Board (IRB)
3. Explain what is and is not Protected Health Information (PHI)
4. Enumerate the 5 situations when a Health Care Providing Entity can disclose PHI to an investigator for the purpose of research
5. Explain what is meant by the de-identification of research data and give examples of identifying data
6. Identify U.S. regulatory policies that address requirements related to animal use in research
7. Explain investigator responsibilities related to use of animals for research

Bioethics

"Bioethics is the study of ethical, social, and legal issues that arise in biomedicine and biomedical research. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics, which addresses ethical issues in public health."

NIH National Institute of Environmental Health Sciences

http://www.niehs.nih.gov/research/resources/bioethics/index.cfm

Institutional Review Board

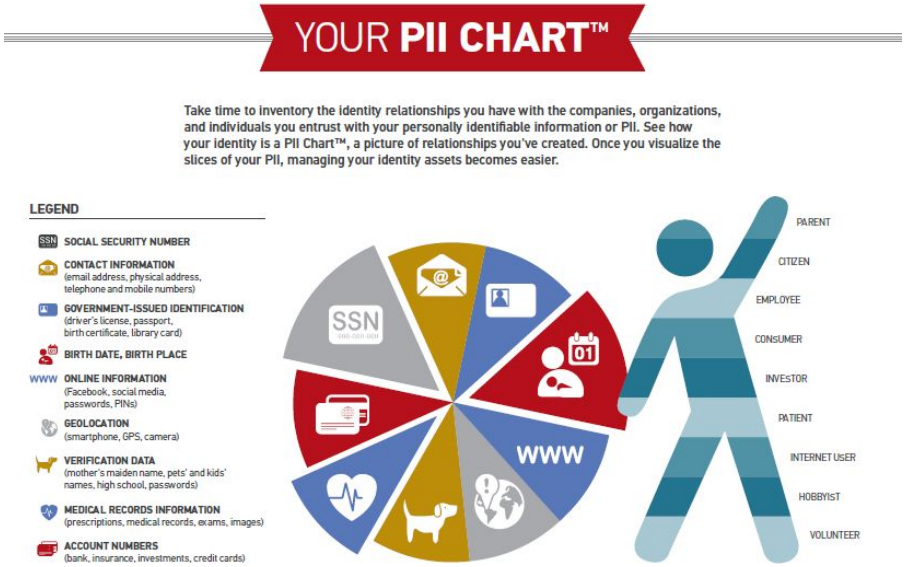
- Institutional review boards (IRB) approve, monitor, and review biomedical and behavioral research involving humans
- IRB assures that appropriate steps are taken to protect the rights and welfare of humans participating as subjects in a research study
- Protocol review assesses the ethics of the research and its methods, promotes fully informed and voluntary participation by prospective subjects capable of making such choices, and seeks to maximize the safety of subjects

Protected Health Information

Protected Health Information (PHI) is any information about health status, provision of health care, or payment for health care that is collected and can be linked to a specific individual

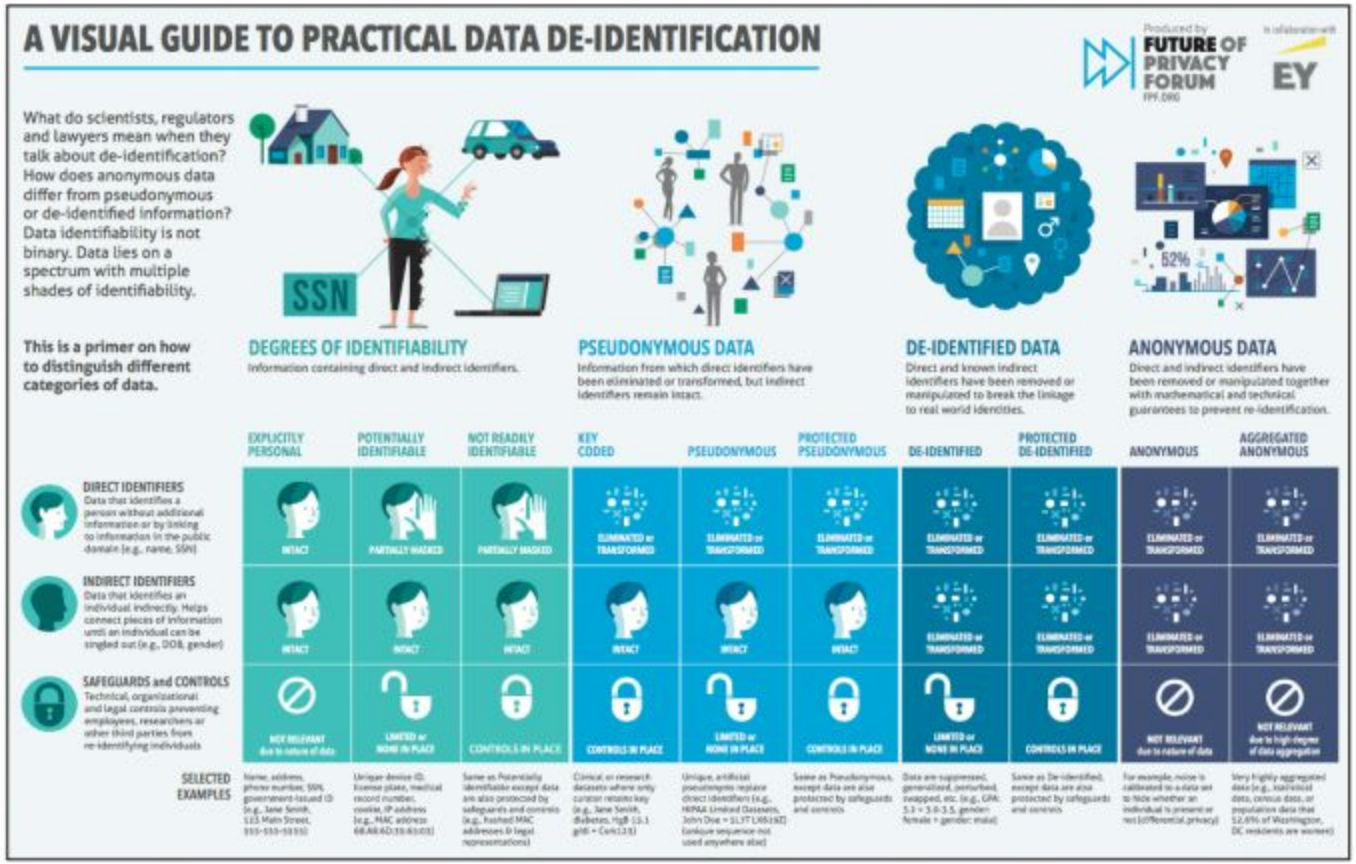
- Includes any part of a patient's medical record or payment history
- Must go through de-identification before researchers share the dataset publicly
- When researchers remove PHI from a dataset they do so in an attempt to preserve privacy for research participants

Personally Identifiable Information (PII) is any data that could potentially distinguish one person from another



Health Insurance Portability and Accountability Act

- HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs
- Administrative Simplification provisions requires the establishment of national standards for electronic health care transactions and national identifiers for providers, health insurance plans, and employers



Polonetsky, Jules, Omer Tene, and Kelsey Finch. 2016. “PII, cookies, and de-ID: Shades of gray.” *Privacy Tech Blog* April 25. https://iapp.org/news/a/pii-cookies-and-de-id-shades-of-gray

Ethical Considerations for Sharing

- Has your IRB approved this (or given you a waiver)?
- Has the appropriate consent from subject been obtained?
- Has the data set had all personal identifiers removed?
- Are you using someone else's data?
- Have you obtained appropriate permissions?
- Are you responsibly using and citing others' data?

Institutional Animal Care and Use Committees

- Most research involving laboratory animals is funded by the National Institutes of Health
- The NIH Office of Laboratory Animal Welfare (OLAW) develops policies that describe the role of IACUCs
- Every institution that uses animals for federally funded laboratory research must have an IACUC
- Each local IACUC reviews research protocols and conducts evaluations of the institution's animal care and use

Module 6: Biomedical Ethics Readings & Resources

Aufderheide, Patricia. 2016. "Does this have to go through the IRB?" *The Chronicle of Higher Education* August 18. http://www.chronicle.com/article/Does-This-Have-to-Go/237476

Oscar A. Zarate, Julia Green Brody, Phil Brown, Mónica D. RamírezAndreotta, Laura Perovich, and Jacob Matz. 2012. "Balancing Benefits and Risks of Immortal Data: Participants' Views of Open Consent in the Personal Genome Project." *Public Health Genomics* 15(2): 106-114. https://dx.doi.org/10.1159%2F000334718

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U.S. Department of Health and Human Services. *Guidance regarding methods for the de-identification of protected health information in accordance with the HIPAA privacy rule* http://www.hhs.gov/hipaa/for-professionals/privacy/special-topics/de-identification/index.html